

Mengdie Huang (Maggie)

Postdoctoral Research Associate, Purdue University

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RESEARCH AREAS

My research focuses on the security and robustness of ML models in critical domains. My overarching goal is to build resilient, trustworthy AI architectures by advancing core research thrusts:

- **Certified Robustness and Vulnerability Assessment:** Scalable certification for provable adversarial robustness guarantees, and realizable evasion attacks to expose vulnerabilities across network and image domains.
- **Robust Transfer Learning and Downstream Adaptation:** Parameter-efficient training and representation distillation for robust downstream adaptation of foundation models against cross-domain threats.
- **Security of Multimodal Foundation Models:** Hardening vision-language models against adversarial attacks and data extraction, with a focus on membership inference and robust VLM adaptation.
- **Adversarial Robustness in Distributed AI:** Securing federated and decentralized learning against evasion attacks and developing self-evolving defense mechanisms for distributed architectures.

EDUCATION

Purdue University , West Lafayette, IN <i>Visiting Ph.D. in Computer Science</i> (Advisor: Dr. Elisa Bertino)	Sep 2022 – Jun 2025
Xidian University , Xi'an, China <i>Ph.D. in Cyberspace Security</i> (Advisor: Dr. Xiaofeng Chen)	Sep 2019 – Jun 2025
Communication University of China , Beijing, China <i>Master in Electronics and Communication Engineering</i> <i>Bachelor in Radio and Television Editing</i>	Sep 2017 – Jun 2019
Huai'an University , Huai'an, China <i>Bachelor in Communication Engineering</i>	Sep 2013 – Jun 2017

WORK EXPERIENCE

Postdoctoral Research Associate , Purdue University, West Lafayette, IN <i>Advisors:</i> Dr. Elisa Bertino and Dr. Ninghui Li <i>Research:</i> Robustness of large language and vision foundation models.	Jul 2025 – Present
Visiting Scholar (Ph.D. Student) , Purdue University, West Lafayette, IN <i>Advisor:</i> Dr. Elisa Bertino <i>Research:</i> Robustness of deep learning-based network traffic analysis systems.	Sep 2022 – Jun 2025
Research Assistant , Xidian University <i>Advisor:</i> Dr. Xiaofeng Chen <i>Research:</i> Robustness of deep learning-based image classification systems.	Sep 2019 – Aug 2022
Research Intern , National Radio and Television Administration <i>Research:</i> Secure video coding standards.	May 2019 – Jul 2019

PUBLICATIONS

* *Equal contribution* † *Mentored student* ✉ *Corresponding author*

Journal Papers

- [1] TDSC 2026 TriXfer: Triple-Cross Transfer Attack on Vision Transformer-Adapted Downstream Classifiers. **Mengdie Huang**, Yingjun Lin[†], Ninghui Li, Xiaofeng Chen, Elisa Bertino[✉].
In *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026.

- [2] TDSC 2025 CARD: Robustness-Preserving Transfer Learning for Network Intrusion Detection via Contrastive Adversarial Representation Distillation.
Mengdie Huang, Yingjun Lin[†], Ninghui Li, Xiaofeng Chen, Elisa Bertino[✉].
In *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2025, pp. 5134-5151.
- [3] TOPS 2025 Dimensional Robustness Certification for Deep Neural Networks in Network Intrusion Detection Systems.
Mengdie Huang, Yingjun Lin[†], Xiaofeng Chen, Elisa Bertino[✉].
In *ACM Transactions on Privacy and Security (TOPS)*, 2025, pp. 1-33.
- [4] TIOT 2024 ARIoTEDef: Adversarially Robust IoT Early Defense System based on Self-evolution against Multi-step Attacks.
Mengdie Huang[✉], Hyunwoo Lee, Ashish Kundu, Xiaofeng Chen, Anand Mudgerikar, Ninghui Li, Elisa Bertino.
In *ACM Transactions on Internet of Things (TIOT)*, 2024, 5(3): 1-34.
- [5] JIT 2019 Sparse Selective Encryption for HEVC 4K Video Using Spatial Error Spread.
Mengdie Huang, Cheng Yang[✉], Hao Li, Jian Shen.
In *Journal of Internet Technology (JIT)*, 2019, 20(5): 1589-1600.

Conference Papers

- [1] TrustCom 2024 MARS: Robustness Certification for Deep Network Intrusion Detectors via Multi-Order Adaptive Randomized Smoothing.
Mengdie Huang, Yingjun Lin[†], Xiaofeng Chen[✉], Elisa Bertino.
In *IEEE International Conference on Trust, Security, and Privacy in Computing and Communications (TrustCom)*, 2024, pp. 767-774. (**Best Paper**)
- [2] AsiaCCS 2023 Boost Off/On-Manifold Adversarial Robustness for Deep Learning with Latent Representation Mixup.
Mengdie Huang, Yi Xie, Xiaofeng Chen[✉], Jin Li, Changyu Dong, Zheli Liu, Willy Susilo.
In *ACM Asia Conference on Computer and Communications Security (AsiaCCS)*, 2023, pp. 716-730.
- [3] CIC 2023 A Pro-Active Defense Framework for IoT Systems.
Elisa Bertino[✉], Hyunwoo Lee, **Mengdie Huang**, Charalampos Katsis, Zilin Shen, Bruno Ribeiro, Daniel De Mello, Ashish Kundu.
In *IEEE International Conference on Collaboration and Internet Computing (CIC)*, 2023, pp. 125-132.
- [4] ESORICS 2022 GAME: Generative-based Adaptive Model Extraction Attack.
Yi Xie, **Mengdie Huang**, Xiaoyu Zhang, Changyu Dong, Willy Susilo, Xiaofeng Chen[✉].
In *European Symposium on Research in Computer Security (ESORICS)*, 2022, pp. 570-588.
- [5] ICCT 2018 Selective Encryption of H.264/AVC based on Block Weight Model.
Mengdie Huang, Cheng Yang[✉], Yuan Zhang.
In *IEEE International Conference on Communication Technology (ICCT)*, 2018, pp. 1368-1373.
- [6] ICCA 2016 An Effective Scheme for Provable Data Possession.
Shanyue Bu[✉], **Mengdie Huang**, Kun Yu.
In *International Conference on Intelligent Control and Computer Application (ICCA)*, 2016, pp. 344-348.

Papers Under Submission

- [1] 2027 Local Knowledge for Global Impact: Evasion Attack Against Vertical Federated Learning.
Chen Peng^{*†}, **Mengdie Huang**^{*✉}, Ninghui Li, Elisa Bertino. (* joint first authors)
- [2] 2027 GRAFT: Adversarially Robust Transfer Learning for Graphs Under Structural Data Scarcity.
Mir Imtiaz Mostafiz^{*†}, **Mengdie Huang**^{*✉}, Imtiaz Karim, Elisa Bertino. (* joint first authors)

- [3] 2027 Local Nonlinear Substitution for Bounding Activation Functions.
Zhongkui Ma, **Mengdie Huang**, Guangdong Bai[✉], Elisa Bertino, Zihan Wang.
- [4] 2027 CARE: Robust Transformer-based Network Intrusion Detection via Consistency-aware Attention-Representation Enhancement.
MD Shamsul Kaonain^{*†}, **Mengdie Huang**^{*✉}, Imtiaz Karim, Elisa Bertino. (* joint first authors)
- [5] 2027 EntroDelta: An Error-Bounded, Data-Free Compression Framework for Massive Model Storage.
Yani Liu[†], Feng Zhang[✉], Yu Zhang, **Mengdie Huang**, Elisa Bertino, Xiaoyong Du.

Books

- [1] Book 2024 Searchable Encryption.
Xiaofeng Chen[✉], **Mengdie Huang**, etc.
In *Science Press, China Science Publishing & Media Co., Ltd.*, 2024.
- [2] Book 2023 Cloud Computing Security (2nd Edition).
Xiaofeng Chen[✉], **Mengdie Huang**, etc.
In *Science Press, China Science Publishing & Media Co., Ltd.*, 2023.

GRANT WRITING EXPERIENCE

Core Author of the NSF proposal , Purdue – West Lafayette, IN <i>Model Robustness, Ownership, and Privacy Preserving in Transfer learning</i>	Sep 2026
Author of the Amazon research proposal , Purdue – West Lafayette, IN <i>Formal and Drift-Aware Containment of Agents in Regulated Systems</i>	May 2026
Core Author of the Cisco research proposal , Purdue – West Lafayette, IN <i>Detection of GenAI Generated Malware Variants and Sandbox Evasion using GenAI</i> Awarded \$150,000 by Cisco Research	Mar 2023

INVITED TALKS

CS 59200-AAS, Department of Computer Science, Purdue University <i>Robustness in Machine Learning – From Non-Adversarial Robustness to LLM Safety</i>	Feb 2026
CS 59200-AAS, Department of Computer Science, Purdue University <i>Robustness in Machine Learning – From Adversarial Robustness to Certified Robustness</i>	Feb 2026
SPARKS Seminar, NSF AI-EDGE Institute <i>Robustness Certification for Deep Neural Networks in Network Intrusion Detection Systems</i>	Dec 2025
School of AI and Computer Science, Shaanxi Normal University <i>AI Security: Adversarial Attacks and Robustness in Deep Learning</i>	Oct 2025

TEACHING

Teaching Assistant , Xidian University <i>Course: Distributed Computing</i> <i>Instructor: Dr. Mirosław Kutylowski, School of Cyber Engineering</i>	Jul 2021 – Jul 2022
Teaching Assistant , Xidian University <i>Course: Probability Theory and Mathematical Statistics</i> <i>Instructor: Dr. Jianfeng Wang, School of Cyber Engineering</i>	Sep 2020 – Jan 2021
Reading Group Instructor , Communication University of China <i>Topic: Principles and Practices of H.264/H.265 Secure Video Coding</i> <i>Format: Organized and led a reading group for undergraduate students</i>	Sep 2017 – Jun 2019

MENTORING

Chen Peng , Ph.D. Student, Purdue University <i>Research: Adversarial machine learning in multimodal domain.</i>	Feb 2026 – Present
Md Shamsul Kaonain , Ph.D. Student, Purdue University <i>Research: Network Traffic Analysis with deep learning.</i>	Jun 2025 – Present
Yani Liu , Visiting Ph.D. Student, Purdue University <i>Research: Data compression in machine learning.</i>	Mar 2025 – Mar 2026
Mir Imtiaz Mostafiz , Ph.D. Student, Purdue University <i>Research: Robustness of graph neural networks.</i>	Jan 2025 – Present
Yuchen Jason Hu , Undergraduate Student, Purdue University <i>Research: Robust Transfer Learning.</i>	Feb 2024 – Present
Yingjun Link Lin , Undergraduate Student, Purdue University <i>Research: Adversarial attack and robustness in deep learning.</i>	Nov 2022 – Dec 2023
Yundong Liu , PhD student, Xidian University <i>Research: Network Traffic Analysis.</i>	Sep 2022 - Jun 2025
Wei Li , Undergraduate student, Xidian University <i>Research: Adversarial attack with Fast Gradient Sign Method.</i>	Dec 2019 - Jun 2020
Fengdong Li , Undergraduate student, Communication University of China <i>Research: Selective Encryption for HEVC Video.</i>	Sep 2018 - Jun 2019
Yujie Wang , Undergraduate student, Communication University of China <i>Research: Selective Encryption for HEVC Video.</i>	Sep 2018 - Jun 2019

PROFESSIONAL SERVICE

Journal Services

Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)	2025 – Present
Reviewer, IEEE Transactions on Networking (TNET)	2025 – Present
Reviewer, IEEE Transactions on Parallel and Distributed Systems (TPDS)	2025 – Present
Reviewer, IEEE Transactions on Information Forensics & Security (TIFS)	2024 – Present
Reviewer, IEEE Transactions on Neural Networks and Learning Systems (TNNLS)	2024 – Present
Reviewer, ACM Transactions on Multimedia Computing, Communications, Applications (TOMM)	2024 – Present
Reviewer, Computers and Electrical Engineering (CEE)	2024 – Present
Reviewer, Computing and Informatics (CAI)	2024 – Present
Reviewer, American Journal of Artificial Intelligence (AJAI)	2024 – Present
Reviewer, ACM Computing Surveys	2023 – Present
Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)	2022 – Present
Reviewer, Telecommunication Systems (TELS)	2021 – Present
Reviewer, Computer Standards & Interfaces (CSI)	2021 – Present
Reviewer, Connection Science	2020 – Present
Reviewer, IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)	2019 – Present

Conference Services

PC Member, IEEE International Conference on Parallel and Distributed Systems (ICPADS)	2026
Artifact Evaluation Committee, ACM Conference on Computer and Communications Security (CCS)	2025, 2026
Artifact Evaluation Committee, Network and Distributed System Security Symposium (NDSS)	2026
Reviewer, International Conference on Data Security and Privacy Protection (DSPP)	2024, 2025
Reviewer, Annual Computer Security Applications Conference (ACSAC)	2024
Reviewer, European Symposium on Research in Computer Security (ESORICS)	2021, 2024, 2025

Reviewer, ACM Conference on Data and Application Security and Privacy (CODASPY)	2023, 2024, 2025
Reviewer, International Conference on Machine Learning for Cyber Security (ML4CS)	2022
Reviewer, Information Security Conference (ISC)	2022
Reviewer, International Conference on Information and Communications Security (ICICS)	2021
Sub-Reviewer, IEEE Symposium on Security and Privacy (S&P)	2025, 2026
Sub-Reviewer, ACM Conference on Computer and Communications Security (CCS)	2025
Sub-Reviewer, USENIX Security Symposium (USENIX Security)	2026

HONORS & AWARDS

Awarded Peer Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)	2025
Awarded Peer Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)	2025
Best Paper Award, IEEE International Conference on Trust, Security, and Privacy in Computing and Communications (TrustCom)	2024
Merit Doctoral Student, Xidian University	2022
Doctoral Academic Scholarship, Xidian University	2019 – 2022
Merit Graduate Student, Communication University of China	2019
Graduate Academic Scholarship, Communication University of China	2017 – 2019
Merit Undergraduate Student, Huai'an University	2017
Undergraduate Academic Scholarship, Huai'an University	2013 – 2017

REFERENCES

Dr. Elisa Bertino, Department of Computer Science, West Lafayette, IN	Purdue University
Dr. Ninghui Li, Department of Computer Science, West Lafayette, IN	Purdue University
Dr. Xiaofeng Chen, School of Cyber Engineering, Xi'an, China	Xidian University